Chapter 1

Introduction: Deadjectival verb formation in Indo-European and beyond

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This volume presents a collection of papers discussing the morphological, syntactic and semantic properties of deadjectival verbs, defined here as verbs formed from property concept roots or from adjectival stems. Although the focus is on older Indo-European (IE) languages, the volume also includes insights from a theoretical perspective using data from modern IE as well as non-IE languages. This chapter serves as an introduction to the volume and provides a precise definition of deadjectival verb formation and an overview over the current state of the art both from the perspective of general and of historical linguistics. We begin with a discussion of different strategies for lexicalizing property concepts, which cross-linguistically constitute the core of the morphosyntactic class "adjective" and various diachronic developments this class can undergo, moving on to describing two types of deadjectival verb formation (root- vs. stem-derivation) and their differentiation from non-deadjectival change-of-state verbs. Subsequently, we introduce the "Caland system" and its implications for the reconstruction of property concept adjectives and the verbal system of the Indo-European proto-language, followed by an outline of recent advances and open questions in the reconstruction of verbal derivational morphology. We conclude with a short summary of each of the respective chapters collected in this volume, a general conclusion, and an outline of open issues.

1 Introduction

1.1 Background: Adjective classes

The papers collected in this volume discuss the morphology, syntax, and semantics of deadjectival verbs and related formations, with a focus on the older Indo-European (IE) languages. The goal of this chapter is to provide a definition and

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overview of the current research on deadjectival verbs and the expression of property concepts in general, with a focus on theoretical morphosemantic studies on the one hand and recent advances in the reconstruction of property concept-associated lexical classes in Indo-European on the other. Ultimately, our goal is to show that while these two strands of literature are often blissfully ignorant of one another, each could profit from a serious engagement with the evidence discussed in the other and the conclusions drawn from this evidence – especially since these conclusions converge more often than not. We also point out some promising avenues for future research in the domain of Indo-European derivational morphology and morphological reconstruction more generally.

Before discussing deadjectival verbs, we need to give a definition of what we mean, which in turn hinges on the definition of "adjective". The literature on this enigmatic lexical category is vast and we cannot do it justice here (see Fábregas & Marín 2017; Panagiotidis & Mitrović 2022; Beck 2023 for recent overviews). For our purposes, we focus on the adjectival classes in ex. (1), which pick out a syntactically and morphologically clearly defined class in the older Indo-European languages: syntactically, members of this class are prototypical noun modifiers (Croft 1991; Alfieri 2014) and morphologically, they are formed with specifically class-defining (adjectival) stem-forming morphology and agree with their head noun.[1] We illustrate these with examples from English and selected older IE languages; note that most of the adjectival suffixes in these examples can be used in more than one class.

(1) Three types of adjectives in Indo-European
  a. Property concept adjectives[2] (e.g., Engl. *tall*, *smart*, *red*, *soft*):
    i. Vedic Sanskrit: *náva-* 'new', *gurú-* 'heavy', *bṛhánt-* 'high, great', *pṛthú-* 'broad', etc.
    ii. Ancient Greek: *kratús* 'strong', *néos* 'young, new', *eruthrós* 'red', etc.
    iii. Latin: *gravis* 'heavy', *clārus* 'clear, bright', *ruber* 'red', etc.
  b. Relational (possessive, genitival, etc.) adjectives (e.g., Engl. *wood-en*, *water-y*, *father-ly*):
    i. Vedic Sanskrit: *aśv-ín-* 'having horses', *āyas-á-* '(made) of metal', *paśu-mát-* 'consisting of cattle', etc.

[1] We cannot discuss other adjectival categories such as participles here for reasons of space; on participles in Indo-European see, e.g., Lowe 2015; Fellner & Grestenberger 2018; Grestenberger 2020.

[2] Also called qualitative (or qualifying) adjectives in the literature.

ii. Ancient Greek: *khrús-eos* 'golden', *odunē-rós* 'painful', *rhodó-eis* 'rosy; of roses', etc.

iii. Latin: *barbā-tus* 'bearded', *patr-ius* 'of the father, fatherly', *argent-eus* 'silvery, of silver', *mar-īnus* 'of the sea, marine', etc.

c. (De)verbal adjectives (e.g., Engl. *wash-able*, *break-able*):

i. Vedic Sanskrit: *kām-ín-* 'desiring', *cá-kr-i-* 'making', *krīḍ-ú-* 'playing', etc.

ii. Ancient Greek: *tom-ós* 'cutting; sharp', *do-tós* 'granted, given', *main-ás* 'raving', etc.

iii. Latin: *trem-ulus* 'trembling', *fīd-us* 'trusting', *aud-āx* 'daring', *amā-bilis* 'lovable', etc.

The classes in (1b–1c) do not usually serve as the input to deadjectival verb formation (though see Sections 2.2 and 3.4 for some circumscribed exceptions to this generalization), so we focus on (1) in the following.

1.2 The semantics & lexicalization of property concepts

The core problem of the definition of "adjective" (and hence "deadjectival") centers on the nature of property concepts (PCs) as defined by Dixon (1982) and illustrated in Table 1.

Table 1: Dixon's property concepts (Dixon 1982: 16; Rau 2009a: 79)

DIMENSION	big, small, long, tall, short, etc.
PHYSICAL PROPERTY	hard, soft, heavy, wet, rough, strong, etc.
COLORS	black, white, red, etc.
SPEED	fast, quick, slow, etc.
AGE	new, young, old, etc.
VALUE	good, bad, lovely, atrocious, perfect, proper, etc.
HUMAN PROPENSITY	jealous, happy, kind, clever, generous, cruel, etc.

As is well known from the typological literature (Dixon 1982, 2004; Thompson 1989, Alfieri 2014, a.m.o.), the lexicalization of PCs differs cross-linguistically in terms of category: Some languages express these concepts as "primary" (root-derived or monomorphemic) adjectives, as illustrated in (2a), while others use a "verbal" strategy, that is, they lexicalize these roots in the same way as eventive ones, namely with verbal (including subject agreement) morphology, cf. (2b). A third strategy that has received less attention but which is highly relevant for the problem of PCs in Indo-European is the lexicalization of PCs as nouns, illustrated in (2c). That "quiet" in (2c) is a noun is suggested by the fact that it takes its own nominal class marker (class 5) rather than agreeing with the nominal class of the subject (class 19).

(2) Cross-linguistic variation in the lexicalization of PCs (Hanink & Koontz-Garboden 2025: 1)

a. I'm **happy**.

b. ma:-č m-**ahan**-kəm
   2-SBJ 2-**be.good**-INCMPL
   'You're good.' Yavapai (Yuman), Kendall 1975: 90

c. hí-nuní híí hí yé li-**mugɛ̂**
   19-bird 19.that 19.SUB be 5-**quiet**
   'That bird is quiet.' Basaá (Bantu), Jenks et al. 2018: 650

Cross-cutting the distinction in terms of lexical category is the difference between what Francez & Koontz-Garboden (2015, 2017) term the "predicative" and the "possessive" strategy: While the predicative strategy makes use of canonical predication devices available in a given language (such as the copula), (3a), the possessive strategy uses possessive morphology to express property concepts, (3b).

(3) Predicative vs. possessive lexicalization of property concepts (Francez & Koontz-Garboden 2017: 22)

a. Pierre is hungry. (English)

b. Pierre a faim. (French)
   Pierre has hunger

Based on this observation, Francez and Koontz-Garboden formulate the generalization that property concept lexemes that use possessive predication denote qualities while property concept lexemes that use non-possessive predication characterize individuals, as in (3a) – The Lexical Semantic Variation Hypothesis (Francez & Koontz-Garboden 2017: 77; but see below for an important diachronic qualification of this generalization). They derive this generalization from the differing denotation of property concepts depending on whether they are lexicalized as nouns or as adjectives: While nouns can denote either qualities or characterize individuals, adjectives can only characterize individuals, but cannot denote qualities. Concretely, this means that (4a) can never have the same meaning as (4b), namely that of expressing the modification of the denotation of the noun.[3] In the same vein, the hypothetical deadjectival verb in (4c) can be individual-characterizing like (4b) meaning "Indra is/becomes strong", but not quality-denoting (Francez & Koontz-Garboden 2017: 83–84). We discuss some equivalents of (4c) in Section 3.3.

(4) a. Indra is strength
    b. Indra is strong
    c. Indra strong-s

While (3b) as such is not used in the older IE languages (though see the discussion of (5) below), (3a = 4b) is amply attested, and with morphology that is actually called "possessive" because it is also used to form denominal relational adjectives from substantives in the older IE languages (though not all languages preserve both functions), e.g., (proterokinetic) *-*u*-, *-*(e/o)nt*-, *-*u̯ont*-, *-*ro*-, *-*mo*-, *-*no*-, etc. (cf. Rau 2009a: 77–78). Examples for the denominal-possessive vs. the property concept root-derived use of these are given in Table 2.

Table 2: Possessive and PC adjectives in Indo-European

Suffix	a. Possessive adj.		b. PC adjective	
-(e/o)nt*-	Hitt.	*perun*-***ant***- 'rocky'	Ved.	*br̥h*-***ánt***- 'high'
base:		*peruna*- 'rock'		√*br̥h* 'high'
*-*ro*-	Ved.	*aṃhu*-***rá***- 'constricted'	AG	*eruth*-***ró***-*s* 'red'
base:		*aṃhu*- 'constriction'		√*ereuth*-/*eruth*- 'red'
*-*u*-	Ved.	*āy*-***ú***- 'lively'	OAv.	*pərəθ*-***u***- 'broad'
base:		*ā́y*-*u*- n. 'life (force)'		√*pərəθ* 'broad'
*-*o*-	Ved.	*pīvas*-***á***- '(having) fat'	Ved.	*śvet*-***á***- 'bright'
base:		*pī́vas*- 'fat' n.		√*śvit* 'bright'

In terms of diachrony, this functional overlap could be taken to suggest that the PC adjectives in column b. were originally *denominal* adjectives derived from quality-denoting nouns. Hence one way of creating individual-characterizing adjectives like (4b) is by adding possessive morphology to the quality nominal, (5a),

[3] It *can* mean "Indra represents strength" or something similar, of course.

or morphology with broadly “instrumental” meaning (e.g., prepositions meaning “with” or, in the case of the older IE languages, instrumental case endings, cf. Section 3.2).

(5) a. Indra is strength-POSS
    b. Indra is with strength/strength-INSTR

Under this view, the b. examples in Table 2 actually instantiate strategy (5a), though whether this is only true diachronically or also synchronically is to some extent a matter of analysis that also touches on the problem of root denotation: are PC roots in Indo-European adjectival in the sense of adnominal/stative (as argued for Proto-Indo-European by, e.g., Nussbaum 1976, 2021a, 2022), quality-denoting (as argued for PC roots in Ulwa by Koontz-Garboden & Francez 2010, Francez & Koontz-Garboden 2017) and hence implicitly nominal (in the sense of “substantival”), or verbal[4] in the sense that they behave in the same way as eventive and other types of stative roots (see Rau, this volume)?

The case of Ulwa, an endangered Misumalpan language spoken in Nicaragua, is especially relevant in this context. In Ulwa, like in the examples in Table 2, the same possessive morphology, namely a suffix *-ka*, is found on possessed nouns, where it marks a 3sg. possessor, (6a), and on PC roots both in predicative, (6b), and in attributive/adnominal position, (6c).

(6) Possessive predication of PCs in Ulwa (Koontz-Garboden & Francez 2010: 200; Francez & Koontz-Garboden 2017: 31)

a. Alberto pan-**ka**
   Alberto stick-3SG.POSS
   ‘Alberto’s stick’

b. yang as-ki-na minisih-**ka**.
   1SG shirt-<1SG.POSS> dirty-3SG.POSS
   ‘My shirt is dirty.’

[4] As implicitly assumed in LIV^2, where most of the reconstructable Indo-European PC roots are listed as forming primary, i.e., root-derived verbs, e.g., **ḱu̯ei̯t-* ‘light up brightly’, LIV^2 340; **leu̯k-* ‘become bright, light’, LIV^2 418–419; **peh*$_2$*ĝ-* ‘become stiff, fixed’, LIV^2 461; **h*$_1$*reu̯d*^{h}- ‘redden’, LIV^2 508–509; **su̯eh*$_2$*d-* ‘become tasty, sweet’, LIV^2 606–607; **tep-* ‘be warm, hot’, LIV^2 629–630; etc. That PCs in Proto-Indo-European (PIE) could be verbal is also explicitly stated by Balles (2009: 8) (“it was obviously possible in PIE to encode “adjectival” concepts by primary verbs”) and argued for by Rau (2009a, 2013, 2016, this volume) and Bozzone (2016), cf. Section 2.2.

In addition to forming primary verbs, PCs also share a number of other derivational properties with eventive roots in (P)IE, such as the formation of root noun and “*tómos*-type” abstracts (Alan Nussbaum, p.c.; see also Nussbaum 2022 on the properties of “adjectival roots” in IE).

c. Al adah-**ka** as tal-ikda.
man short-3SG.POSS INDEF see-1SG.PAST
'I saw a short man.'

The parallel is not exact (the Ulwa marker is an agreement marker, whereas the suffixes in Table 2 are primary categorizing and/or derivational suffixes which cannot be used in "possessive genitive" contexts such as (6a)), but it does not seem trivial that the adjectival equivalents of (6b–6c) in Table 2 use morphology that also forms relational (and very often specifically possessive) adjectives. In fact, as Hanink & Koontz-Garboden (2025) argue based on Menon & Pancheva (2014), there is cross-linguistic evidence that "categorizers of property concept roots are tied up with possession in the creation of nominal, verbal, and adjectival property concept words" (Hanink & Koontz-Garboden 2025: 5). In these cases, the categorizer lexically categorizes the root and at the same time introduces a possessive relation: Possessive categorizers take a property P (a PC root) and return the set of individuals x that are in possession of this property (Hanink & Koontz-Garboden 2025: 6).

An interesting diachronic problem arises from this proposal that remains to be explored, namely how this type of root-derived possessive categorization is related to possessive morphology that selects already categorized nominal bases, as in column a. of Table 2. At least in the older IE languages, there is ample evidence that the latter can give rise to root-derived possessive categorizers through a reanalysis of the categorized base as an uncategorized root (or a root with a zero categorizer), cf. Grestenberger (2020, 2021), Grestenberger et al. (2023). We discuss this further in Section 3.1. First, we want to emphasize that the existence of forms like those in Table 2, that is, possessive and PC-related adjectival morphology, which can be securely reconstructed for PIE, provides a strong argument against the view that PIE did not have adjectives as a distinct morphosyntactic class, as proposed by Balles (2006) and more recently in a different form by Alfieri (2016). Balles argues that adjectival/PC root-derived nominals, especially zero-derived "root nouns" and *i*-stems, were categorially underspecified and could be used both as adjectives (e.g., in attributive modification contexts) and as nouns (e.g., as arguments). But the existence of a productive process of zero derivation of substantives from adjectives (not just PC adjectives) does not mean that the categorial distinction is moot, since the substantivizations are morphosemantically clearly secondary, i.e., based on the adjectival form and meaning (e.g., Lat. *datum* 'given' (neuter adj.) → 'what is given, gift', neuter substantive). The existence of denominal possessive adjectival suffixes in turn suggests that the categorial distinction was both derivationally and syntactically relevant, as Balles herself admits (2006: 286). That specifically PC related adjectival morphology can be reconstructed for the oldest layer of the proto-language (as well as the comparative and superlative suffixes *-*i̯os*- and *-*is-tó*- for at least the inner Indo-European[5] stage) is another strong argument against "word class indifference".[6]

Alfieri, on the other hand, has argued in a series of articles (Alfieri 2016, 2019, 2021; Alfieri & Gasbarra 2021) that PIE was an adjective-less language by classifying PC roots as "verbal roots of stative-unaccusative meanings" (Alfieri 2016: 159) or "verbal root[s] meaning a quality" (2021: 314). Adjectives based on such roots are hence treated as verbal adjectives and actually grouped with verbal stem-derived participial forms, though with a different range of functions than the latter. Alfieri's treatment of these forms is primarily based on his interpretation of the Vedic evidence and acknowledges that the Vedic system differs from the Part-of-Speech (PoS) systems of, for example, Ancient Greek and Latin. But even if there were compelling reasons for privileging the Vedic PoS system over that of the other daughter languages for the purposes of reconstruction, this approach ignores the cross-linguistic evidence for a systematic morphosemantic difference between PC roots and other types of CoS-associated roots for which correlates are also found in the older IE languages (see Sections 2.2 and 3.3 for a more detailed discussion) as well as the specifically adjectival morphology associated with the "Caland system" (= PC roots and their derivatives in (Proto-)Indo-European, cf. Section 3.1) and with (denominal) relational adjective formation, all of which can be securely reconstructed for PIE.[7]

Finally, strategy (5b) has also played an important role in the scholarship on PCs in Indo-European in the context of the interpretation of the "*cvi* construction", an Old Indic predicative construction consisting of an indeclinable form of a noun (sometimes described as adverbial in the literature) and a finite form of a copula or light verb, usually *as*- 'be', *bhū*- 'become', and *kr̥*- 'make', with which it is sometimes univerbated (Schindler 1980; Balles 2000, 2006, 2009). The name of the construction, *cvi*, is the label used by the ancient Indian grammarians (e.g., Pāṇini, *Aṣṭādhyāyī* 5.4.50). Some examples of this construction are given in Table 3.

While Old Indic and the Indo-Aryan languages in general are no strangers to periphrastic constructions consisting of a verbal noun + light verb, the problem

[5] I.e., the branches that descended from a common ancestor (≈ Late PIE) after the ancestral languages of the Anatolian and Tocharian branches left the family.

[6] See Meier-Brügger 2014, Rau 2014, Pinault 2018 on the prehistory of the comparative suffix and its role in the Caland system.

[7] See further Acedo-Matellán 2022b for arguments against the existence of "adjective-less languages".

Table 3: Old Indic *cvi* forms (ex. from Balles 2006: 45–46)

krūrá-	‘sore’	*krūrī́ kr̥-*	‘make sore’
náva-	‘young, new’	*navī kr̥-*	‘renew, rejuvenate’
phála-	‘fruit’	*phalī́ kr̥-*	‘winnow, clean’
yúvan-	‘young’	*yuvībhū-*	‘become young’

of the *cvi* construction lies in the interpretation of the *-ī* of the nonfinite part of the construction. The interpretation as an instrumental singular **-i-h₁* of an adjectival abstract noun, effectively a construction of type (5b), is appealing in light of the cross-linguistic evidence of languages using a possessive strategy for lexicalizing property concepts. Given that PIE did not have a HAVE lexeme, it was most likely an S-POSSESSOR and/or S-POSSESSEE language (evidence for both strategies exist in the older IE languages), in which possessive predication was expressed through an intransitive construction in which either the possessor (S-POSSESSOR) or the possessee (S-POSSESSEE) was coded as the subject (Heine 1997, Stassen 2009). With its reconstructed deinstrumental origin, the *cvi* construction would fit the S-POSSESSOR pattern, specifically the COMITATIVE-POSSESSIVE type (Creissels 2023) also seen in, e.g., Hausa, (7), cf. the “*with*-possessive” in Stassen (2009: 54–57).[8]

(7) Hausa (Chadic), Newman 2000: 222 (cit. after Creissels 2023: 38)

Yarȍ	yanȁ	dà	fensìr̃
boy	3SG.M.ICPL	with	pencil

‘The boy has a pencil.’, lit. ‘The boy is with pencil.’

Although there are solid structural parallels to the *cvi* construction in other IE languages that point towards an inherited strategy of possessive predication with instrumental nouns (see Sections 3.2 and 3.3), as well as robust typological parallels, its diachronic connection to the expression of PCs in Indo-European in general and Old Indic in particular awaits further elucidation.[9]

[8] Cf. also Heine (1997: 53–57). The S-POSSESSEE strategy with genitive- and dative-marked possessors also existed in the older IE languages.

[9] The antiquity of the construction has been called into question because it is practically absent in the oldest Vedic text (the Rigveda) and is not specifically associated with inherited PC roots or lexemes (Jamison 2023). Rather, it expresses predicative possession in general, and the derivational basis can be adjectival or nominal (see Balles 2006: 45–75; 151–157 for a detailed discussion), as also seen in the examples in Table 3. In that sense, the *cvi* construction falls

With this introductory background in mind, we can now turn to the question of deadjectival verbs. That is, our focus in the following as well as in the papers in this volume is on the verbal strategy for expressing PCs, i.e., the one illustrated in example (2b) and the hypothetical (4c). However, most of the contributions in this volume are not restricted to the discussion of deadjectival verbs in this strict sense, but also address how such verbs arise diachronically and how they interact with and differ in distribution from the other strategies discussed above. In the next section, we provide a more detailed definition of deadjectival verbs and a discussion of the different types thereof.

2 Deadjectival verbs cross-linguistically

2.1 The name of the game

In the previous section, we have focused on deadjectival verbs formed to roots that express "adjectival meaning", i.e., property concepts (PCs). This is the definition commonly found in the theoretical literature (e.g., Hale & Keyser 2002; Koontz-Garboden 2005, 2014; Bobaljik 2012; Beavers & Koontz-Garboden 2020) and also adopted in some recent Indo-Europeanist literature on the "Caland system" (cf. Section 3.1), e.g., Rau, this volume. Some examples are given in (8a).

There is a second, less salient use of the term that can be paraphrased as in (8b) and essentially defines deadjectival verbs primarily via their morphology, that is, as based on an adjectival *stem* whose adjectival categorizing morphology is contained in the derivative.

(8) Two types of deadjectival verbs

  a. Root-derived verbs from PC roots

    i. Latin

      - *mac-ēre* 'be thin' (Pl.), *mac-ēscere* 'become thin' (Pl.+) ← **mak̂*- or **meh₂k̂*- 'long, thin' vs. adj. *macer*, *mac**r**-* (Nussbaum 1976: 51; NIL 478–479)
      - *rub-ēre* 'be red', *rub-ēscere* 'become red' ← **h₁reu̯dʰ*- 'red' (LIV[2] 508–509) vs. adj. *ruber*, *rub**r**-*

into the general late Old Indic/post-Vedic Sanskrit tendency to increasingly use analytic "light verb" constructions to express complex verb meanings, such as the periphrastic causative, perfect, future, habitual, etc. (Zehnder 2011; Ittzés 2015, 2022; Lowe 2017; Grieco 2023). Note that the structurally related *gúhā bhū-/kr̥-* construction ('become/make hidden/with hiding') is of Rigvedic age, cf. further Section 3.2.

      - *tep-ēre* 'be warm' ← **tep-* 'be warm, hot' (LIV^2 629–630) vs. adj. *tep**id**us*

   ii. Slavic (PSl.)

      - **rŭd-ěti sę*, *rŭdi-*[10] 'grow red, blush' ← **$h_1$reu̯d^h-* 'red' vs. adj. **rŭd**r**ŭ*
      - **top-iti*, *topi-* 'to heat' ← **tep-* 'be warm, hot' (compare Ved. *tāpáyati* and Yav. *tāpaiieiti*, LIV^2 629–630) vs. adj. **tep**l**ŭ*

   iii. Hebrew (Arad 2003: 749; 743)

      - *xizeq* 'strengthen' ← √*xzq* vs. adj. *xazaq*
      - *hišmin* 'grow fat, fatten' ← √*šmn* vs. adj. *šamen*

b. Verbs derived from overtly marked/categorized adjectives

   i. Latin

      - *mac**r**-ēscere* 'become thin' (Hor.+) ← *macer*, *mac**r**-* 'thin'
      - *pig**r**-ēre* 'to be reluctant' (Acc.) ← *piger*, *pig**r**-* 'torpid, inactive' (EDL 464)
      - *crūd**ēl**-ēscere* 'become cruel' (Aug.) ← *crūd**ēl**is* 'cruel' (← *crūdus* 'raw, bloody, rough, cruel')

   ii. Slavic (OCS)

      - *ost**r**iti*, *ostri-* 'sharpen' ← *ost**r**ŭ* 'sharp' (ESJS 601)
      - *ob-lĭg**ŭč**iti*, *ob-lĭgŭči-* 'lighten, make light' ← *lĭg**ŭk**ŭ* 'light' (ESJS 447–448)
      - *sŭ-tep**l**iti*, *sŭ-tepli-* 'to warm, to heat' ← *tep**l**ŭ* 'warm, hot' (ESJS 955)

   iii. Hebrew (Arad 2003: 752, fn. 11)

      - *hitqamc**en*** 'act miserly' ← *qamc**an*** 'miser' (√*qmc*)
      - *hištaxc**en*** 'act arrogantly' ← *šaxc**an*** 'arrogant' (√*šxc*)

While the verbs in both (8a) and (8b) are semantically deadjectival and display meanings expected of deadjectivals ('be X', 'become X', 'make X'), only the latter ones are *morphologically* adjective-derived and contain the stem-forming morphology of the adjectival base. The difference is especially clear when comparing "doublets" like the first of the Latin examples, where the verbal morphology

[10]The citation forms used for Slavic verbs are usually the infinitive and the first person singular, e.g., **rŭždǫ sę*. For ease of comparison (in particular with Lithuanian, where the citation forms are the infinitive and third person), we use the present stem here instead of the first person form. The past forms are omitted.

of *mac-ēscere* in (8a-i) attaches directly to the root, while *mac-**r**-ēscere* in (8b-i) clearly shows the *-r-* of the adjective stem that is kept in the derivative, and the same holds for *pig**r**ēre*. The productivity of the latter strategy of stem-derived formations is confirmed by *crūd**ēl**ēscere* from *crūd**ēl**is*, which is itself a derivative of the adjective *crūdus*.

A similar situation is encountered in the Slavic examples. In (8b-ii) we see adjectives that are derived from typical Indo-European "Caland" roots (*$h_2$ek̂-,[11] *$h_1$leng^{uh}- and *tep-) by using the associated adjectival suffixes *-ro- (which is also the source of the Latin *-r-* in the examples above), *-u- and *-lo-. Those suffixes surface in the respective derived verbs, i.e., they are clearly derived from the adjective rather than from the root. In this way, we get *sŭtep-**l**-iti* as opposed to the root-derived causative **top-iti* in (8a-ii), or Latin *tep-ēre* in (8a-i). Particularly illustrative of this pattern is *oblĭgŭčiti* in (8b-ii): Old **u*-adjectives were regularly extended by the suffix *-ko- in Slavic. This extension can famously be dropped in derivation (e.g., ChSl. *ǫz-iti* 'to narrow' ← *ǫzŭkŭ* 'narrow' < *$h_2$emĝ^h-u-) and in doing so behaves like other Caland suffixes (cf. Majer 2017: §4.2 for a detailed discussion with references). However, the *-č-* of *oblĭgŭčiti* is the reflex of the palatalized *-k-* from *lĭgŭkŭ*, which means that the verb must be derived from the whole synchronic *stem* rather than from the root. Further evidence from the individual IE languages as well as comparative reconstruction suggest that the latter strategy in (8b) is an innovation in the PC context, becoming more and more productive and replacing the root-derived principle in (8a); see, e.g., Watkins (1971) and Nussbaum (1976) for more examples from various Indo-European branches, and Majer (2017) and Reiter (2023) for (Balto-)Slavic.

The contrast between root- and adjective-derived verbs is not restricted to Indo-European languages. In the Hebrew examples in (8a-iii), the root-derived verbs are created by inserting certain vowel combinations into the root consisting of three consonants (and, in the second example, a prefix), the same strategy as is used for the creation of deradical adjectives. On the other hand, (8b-iii) shows verbs derived from the adjectives rather than from the root, as the adjective-building suffix *-an* is retained (albeit with a change of the vowel quality). The "act like" semantics are a manifestation of the broader semantics of stem-derived verbs as opposed to root-derived ones (see Kastner 2020: 194–195 for further examples).

In Indo-European studies, the morphology-oriented use of the term "deadjectival" has been the dominant one so far, i.e., it is used almost exclusively for

[11]But see Nussbaum (2021a: 10) for potential problems of an analysis of PIE *$h_2$ek̂-ro- as historical PC adjective due to the unusual accent of Lith. *aštrùs* and AG *ákros*, which points to *$h_2$ék̂-ro- rather than expected *$h_2$ek̂-ró-.

verbs of type (8b), while (8a) is usually discussed under the label "Caland system" (cf. Section 3.1). Since verbs like those in (8b) are stem-derived, they are traditionally labeled "secondary" as opposed to "primary" formations that are derived directly from the root (cf. above).[12] As a consequence, primary verbs are usually not labeled "deadjectival" (or more broadly "denominative") because they are not deadjectival from a morphological point of view, even if they are semantically. For example, PSl. **rŭděti sę*, *rŭdi-* 'grow red, blush' is usually taken as a primary verb in the literature as it is morphologically deradical and builds an *i*-present as opposed to morphologically unambiguous deadjectivals that form a different type of present stem (e.g., OCS *o-slaběti*, *o-slaběje-* 'become weak' ← *slabŭ* 'weak'; cf. Villanueva Svensson, this volume). Consequently, it is not discussed in the literature on deadjectivals (but see Villanueva Svensson 2021: 275 for an exception). Of course, when analyzing a verbal form, a clear-cut decision on the primary vs. secondary status is sometimes difficult to make (cf., e.g., Boldt, Höfler & Stifter and Sasseville, this volume, or again Villanueva Svensson 2021) and the analysis might differ depending on whether one takes a synchronic or diachronic viewpoint: As is shown by various contributions in this volume, historical secondary formations can at some point be reanalyzed as primary, and a synchronic (re)association of an originally independent derivate with a nominal formation is possible, too.[13] This can lead to a situation where a verb is labeled differently across publications (within one branch as well as across different daughter languages of Indo-European[14]), which makes a comparison somewhat difficult – especially when the criteria for classifying a given example as (synchronically or historically) primary or secondary are not made transparent.

As these examples and the elaborations in Section 1.2 show, an equation of "deadjectival" with "secondary" in the morphological sense of (8b) would exclude an important amount of evidence, especially in the case of the "Caland" phenomenon, as will be further discussed in Section 3.1, and is at odds with the use of the term in theoretical and typological studies. We therefore use the term "deadjectival" in both meanings defined above to encompass both secondary verbs

[12]Because it is common practice in the discipline to set up the root and its meaning as verbal wherever possible (cf. footnote 4), the term "deverbal" or "deverbative" is sometimes inaccurately equated with "deradical" or "primary" in morphological terms. This should be avoided.

[13]Compare Jasanoff (2004: 147–148), who calls the distinction between denominative and deradical in the context of Latin *ē*-verbs "illusory, or at least epiphenomenal" because historically, the ultimate source of the verbal formation itself is denominative, cf. Section 3.2.

[14]E.g., PSl. **rŭděti sę* is equated with Latin *rubēre*, Old Irish *ruidid* 'blush' and OHG *rotēn* 'be reddish, become red', cf. Höfler & Stifter, this volume.

built on overtly marked adjectival stems as well as morphologically primary but semantically deadjectival verbs derived from Property Concept roots (cf. Section 4 for further advantages of this use).

Finally, note that in languages with little or no adjectival categorizing morphology like English, the matter of the distinction between (8a–8b) becomes a serious theoretical problem: Are there verbs which are morphologically deadjectival, but happen to be derived from adjectives in which the adjectival categorizer is zero? Would we expect such verbs to be semantically distinguishable from genuinely root-derived verbs as in (8a)? This also depends on whether we expect verbs of type (8a) to systematically differ semantically from verbs of type (8b), as is suggested *inter alia* by the Hebrew examples and the iteratives discussed in Section 3.4. Further, the problem of root- vs. stem-based derivation links to the question of why some deadjectival CoS verbs are transparently built on the comparative while most others are not, e.g., Engl. *to worsen*, *to lessen*, *to better* vs. *to broaden*, *to clean*, etc., a phenomenon that is found in the older IE languages as well.[15] The same question arises for the verbs of type (8b), which also show overt adjectival (though positive rather than comparative) morphology. We must leave this to future research; in this volume, the contributions by Sasseville, Merritt, Iacobini & De Rosa and Tanza-Kamil treat the complex relationship between verbs that contain overt adjectival morphology (at least historically) and primary/root-derived verbs.

2.2 Two types of change-of-state verbs

In discussing deadjectival CoS verbs, we must moreover be careful to distinguish verbs formed from PC roots from other types of CoS verbs whose base is not adjectival in one of the two senses defined in Section 2.1. Beavers & Koontz-Garboden (2020: 58) observe that "there is a split between two types of change-of-state verbs in terms of whether the entailment of change is introduced solely by the template or an entailment of change is (also) present in the root itself", a difference that can be paraphrased as that between adjectival and eventive (in the broad sense) roots. The first ("adjectival") class that has no entailments of change gives rise to deadjectival CoS verbs, (9). These correspond to the PC states of Dixon (1982) and "Caland" adjectives in the older IE languages (Section 3.1).

[15] Semantic arguments against taking deadjectival degree achievement verbs such as *flatten*, *widen*, *straighten*, etc., from the positive adjective are discussed by Beavers & Koontz-Garboden (2020: 42–45 & fn. 24), etc.; see also Bobaljik 2012: 169–207; Beavers & Koontz-Garboden 2020: 46, fn. 26; Fábregas 2023: 80–90. On the lack of zero-derived deadjectival verbs see Acedo-Matellán 2022a.

The second class of CoS verbs is derived from “states that are the result of some action” (Dixon 1982: 50), (10); cf. also Rau 2013 for a more fine-grained version of this distinction.

(9) Deadjectival CoS verbs (Levin 1993: 245, after Beavers & Koontz-Garboden 2020: 58)
awaken, brighten, broaden, cheapen, coarsen, dampen, darken, deepen, fatten, flatten, freshen, gladden, harden, hasten, heighten, lengthen, lessen, lighten, loosen, moisten, neaten, quicken, ripen, roughen, sharpen, shorten, sicken, slacken, smarten, soften, stiffen, straighten, strengthen, sweeten, tauten, thicken, tighten, toughen, weaken, widen, …

(10) Non-deadjectival CoS verbs (Beavers & Koontz-Garboden 2020: 58)
a. Levin’s (1993: 241) *break* verbs: break, chip, crack, crash, crush, fracture, rip, shatter, smash, snap, splinter, split, tear
b. Levin’s cooking verbs (Levin 1993: 243): bake, barbecue, blanch, boil, braise, broil, charbroil, charcoal-broil, coddle, cook, crisp, deepfry, fry, grill, hardboil, poach, …
c. Verbs of killing (Levin 1993: 230–233): crucify, electrocute, drown, hang, guillotine, …
d. *inter alia*

Crucially, these two classes systematically differ with respect to their morphology and semantics across a wide sample of genetically unrelated languages, as Beavers et al. (2021) show: While the verbs in ex. (9) form both basic and derived statives (= primary and secondary/deverbal adjectives, in Indo-Europeanist terminology), the verbs in ex. (10), or “result roots”, can only form derived statives which always entail a change of state.[16] This difference is illustrated in Table 4 for English (cf. also Rau 2009a: 162–163 on the same contrast in Vedic Sanskrit and Latin).

The same contrast is found in languages that use the verbal strategy (or a mix of strategies) for statives. In Hebrew, for example, both adjectival and result

[16] The “result stative” category in Table 4 and Table 5 conflates different subtypes of (participial) states, e.g., the distinction between target and resultant states of Kratzer (2001), Anagnostopoulou (2003), Anagnostopoulou & Samioti (2014), Alexiadou et al. (2015), etc. What matters for our purposes is that this (composite) category differs systematically from that of the “basic statives”. For a differing view under which “statives” (including underived adjectives) and adjectival passives (≈ Kratzer’s target states) are grouped together to the exclusion of “resultatives” (≈ Kratzer’s resultant states) see Embick (2004).

Table 4: Basic and derived statives in English

Adjectival root	Basic stative	Verb	Result stative
√FLAT	*flat*	*flatten*	*flatten-ed*
√RED	*red*	*redden*	*redden-ed*
Result root	Basic stative	Verb	Result stative
√CRACK	–	*crack*	*crack-ed*
√COOK	–	*cook*	*cook-ed*

statives are categorially *verbal*. While simple statives in Hebrew do not entail a change of state, result statives always do, both for adjectival and for result roots, and they also differ morphologically, cf. the examples in Table 5 (a similar pattern of primary and secondary verbal stative formation is discussed by Tanza-Kamil, this volume, for Akkadian).

Table 5: Basic and derived statives in Hebrew (ex. from Beavers et al. 2021: 457)

Root		Simple stv	Inchoative	Causative	Result stv
Adjectival	√*gdl* 'large'	*gadol*	*gadal*	*higdil*	*mu-gdal*
Result	√*švr* 'break'	–	*ni-šbar*	*šavar*	*šavur*

Given how widespread the morphosemantic differentiation between adjectival and non-adjectival/result CoS verbs is cross-linguistically, this suggests that the (often informal) distinction between "verbal" and "adjectival" roots found in some of the Indo-Europeanist literature (e.g., Nussbaum 2022) is indeed theoretically and typologically valid. In other words, adjectival/PC roots have different combinatorial morphological properties than eventive/result roots, and this reflects their underlying semantic difference. In order to terminologically distinguish between verbs derived from PC vs. eventive/result roots, we therefore use the terms *fientive* (Lat. *fieri* 'become') and *factitive* for, respectively, intransitive and transitive CoS verbs from PC roots and *inchoative* and *causative* for intransitive vs. transitive CoS verbs from eventive/result roots from here on (cf. Villanueva Svensson, this volume). Note, however, that these two groups can also interact and sometimes even share the same verbal morphology (cf. Rau 2013 & this volume; Sasseville, this volume; as well as fn. 4 and Table 8). Moreover, reanalysis of secondary adjectives as primary adjectives, lexicalization, etc., can blur the lines between these two classes for individual lexical items, so that "variation in the interpretation of particular adjectival forms among both simple and derived adjectives is actually expected owing to lexical drift or idiosyncratic semantic factors." (Beavers et al. 2021: 449; cf. also Beavers & Koontz-Garboden 2020: 71–73). For example, the adjective *broken* in (11), also from Beavers et al. (2021: 449) does not entail a change of state, which therefore can be denied. This is unexpected given that *broken* is formed from a result root.

(11) The window is broken but it never broke. It was manufactured that way.

Beavers & Koontz-Garboden (2020: 72) suggest that *broken* could have "undergone lexical drift for some speakers to have a specialized meaning like "not functioning" or "not canonical in a way that suggests defects," which could be an inherent property rather than one derived through a change". In other words, new property concept lexemes can arise through reanalysis or lexical drift, a development that has also been observed in the context of the Caland system in the older IE languages (e.g., Nussbaum 1976, 2021a, 2022; Rau 2009a: 118–126; Alfieri 2016). Reiter (2023: 127–138) discusses some examples in Slavic, in which formations that are traditionally explained as historical participles or resultative deverbal adjectives function as synchronic adjectives with a PC reading that can in turn serve as base for further deadjectival verbs, cf. Table 6. This is also the

Table 6: Slavic deadjectivals from former participles (Reiter 2023: 128; 138). Participial morphemes are bolded.

Base verb	Ptcp. → adj.		Deadjectival verb
PSl. **zelti*[a]	OCS *zel**en**ŭ* 'green'	ChSl.	(*pro-*)*zele**n**ěti* (*sę*), -*ěje-* 'become green'
		Russ.	*zel**en**ét'* 'id.'
		Russ.	*zel**en**ít'* 'make green'
OCS *črŭviti* 'dye red' (← *črŭvĭ* 'worm')	OCS *črŭv(lj)**en**ŭ* 'red'	Cz.	*červ**en**ěti* 'become red'
		Cz.	*červ**en**iti* 'paint red'

[a] cf. Lith. *žélti, žẽlia/želi* 'green, sprout'. Note, however, that for the first example there might also be an alternative explanation since the PIE root *$\hat{g}^h elh_3$- means 'green, yellow, pale', i.e., it has PC semantics. Therefore, the apparent base verb and its "participle" could also be independent derivations.

case for many prototypical PC adjectives in English, e.g., *old*, which goes back to a past participle of the PGmc. verb **alan-* 'grow, bring up' (EDPG 20; Beavers et al. 2021: 449).

All in all, these facts suggest that deadjectival verbs need to be distinguished from non-deadjectival CoS or result verbs, from which they differ semantically (PC vs. result/eventive roots). Morphologically, they are often treated as primary (root-derived) in much of the theoretical and Indo-Europeanist literature, but there are also instances in which they appear to contain overt adjectival stem-forming morphology, and in these cases one must carefully distinguish between instances in which the adjectival morphology has been reanalyzed and essentially acts as verbalizing morphology (cf. Section 3.3) and instances which are derivationally transparent and which raise the question why adjectival morphology is contained in the derived verb and what semantic function (if any) it contributes compositionally.

Since the discussion here and in the following focuses largely on CoS contexts, it is important to point out that deadjectival verbs can also be states and activities. Fábregas (2023) distinguishes between the three broad classes of deadjectival verbs given in ex. (12) in his study of derived verbs in Spanish.

(12) Three types of A > V derivation (Fábregas 2023: 63)
  a. Change of state verbalisation, where the adjectival base defines the set of properties used to evaluate the change undergone by an argument (*claro* 'clear' > *aclarar* 'to make clear')
  b. Activity property verbalisation, where the adjective defines a set of properties exhibited by an argument when performing an event (*holgazán* 'lazy' > *holgazanear* 'to act lazily')
  c. Stative property verbalisation, where the adjective defines the properties exhibited by an argument, and not subject to change or transformation (*transparente* 'transparent' > *transparentar* 'to be transparent')

For reasons of space, we focus on the first class in the following, though see Section 3.4 for a brief discussion of derived activity verbs as in (12b).[17]

Having laid out the theoretical foundations of the topic as well as the definition of deadjectival verbs used in this volume, in the next section we move on to discussing the encoding of PCs in the Indo-European languages in more detail.

[17] See also Oltra-Massuet & Castroviejo 2014 and Acedo-Matellán & Gibert-Sotelo 2022 on this type.

3 Property Concepts and deadjectival verbs in Indo-European

3.1 The "Caland system"

In the older Indo-European languages, property concepts are usually discussed under the label "Caland system", named after Willem Caland (1859–1932), who first observed the peculiar morphological behavior of primary adjectives and their derivatives in Indo-Iranian (Caland 1892, 1893). Caland noticed that primary adjectives, especially those formed with the Indo-Iranian suffixes *-*ra*-, *-*ma*-, and *-*ant*-, seemingly "lost" their adjectival suffix in comparatives, superlatives, and in compounds, often replacing it with a suffix *-*i*- otherwise associated with *nominal* stem formation. This replacement pattern is illustrated in Table 7 with examples from Vedic, Avestan (both Indo-Iranian) and Ancient Greek (AG). Essentially, while comparatives, superlatives, and abstracts from primary ("Caland") adjectives are semantically deadjectival or "secondary", they seem to be morphologically root-derived or "primary" formations.

Table 7: The Caland system in Indo-Iranian and Ancient Greek

	Positive adj.		Comparative	Superlative	Abstract
Ved.	*tig-má-*	'sharp'	*téj-īyaṃs*	*téj-iṣṭha-*	*téj-as-*
	pr̥th-ú-	'broad'	*práth-īyaṃs-*	*práth-iṣṭha-*	*práth-as-*
OAv.	*ug-ra-*	'strong'	*aoj-iiah-*	*aoj-išta-*	*aoj-ah-*
YAv.	*bərəz-ant-*	'high'	*barəz-iiah-*	*barəz-išta-*	*barəz-ah-*
AG	*kūd-rós*, *-nós*	'glorious'	*kūd-íōn*	*kū́d-istos*	*kūd-os*
	krat-ús	'strong'	*kréssōn* (< **krét-jōn*)	*krát-istos*	*krát-os*

Subsequent research has shown that this descriptive replacement of the adjectival morphology with *-*i*- in composition, termed "Caland's Law", is neither confined to Indo-Iranian nor to the suffixes discussed by Caland (see Dell'Oro 2015 for an overview of the research history). Building on earlier works by Wackernagel (1897) and Risch (1974), Nussbaum (1976) showed in his seminal study that the alternating adjective- and abstract-forming suffixes must be conceptualized as forming a complex derivational *system* to roots with "inherently adjectival semantics" (Nussbaum 1976: 6). While some of them are more frequent and therefore "central" to the system, as the aforementioned *-*ro*-, *-*u*-, *-*(e/o)nt*- and

*-*i*- (and, added later, simple thematic *-*o*-, cf. Nussbaum 2021a, 2022),[18] others are less frequently used for forming PC adjectives and therefore "marginal" or "peripheral", like *-*mo*-, *-*no*-, *-*lo*- and *-*to*- (see Rau 2009a: 71–74). A combination of suffixes, like *-*i-no*- or *-*u-mo*- is possible as well. The reason for the synchronic substitution principle is to be found in the derivational base of the respective formations: Since for a number of roots displaying a Caland system a root noun (that is, a noun built deradically with a zero suffix/categorizer) with abstract semantics is either attested or reconstructable, Nussbaum (1976: 100–105) argues that the seemingly primary formations with alternating suffixes are ultimately parallel derivations from such a root noun and consequently (historically) *secondary* formations, as outlined in ex. (13).

(13) Derivation of Caland adjectives from root nouns (Rau 2009a: 129–130)
   a. root noun *$kr\acute{u}h_2$-Ø- 'gore, raw or bloody flesh'
      → adj. *$kruh_2$-ró- 'bloody, gory'
   b. root noun *h_1rud^h-Ø- 'redness'
      → adj. *h_1rud^h-ró- → 'red',
      → adj. *$h_1re/o\underset{\frown}{u}d^h$-ó- 'possessing redness' → 'red'

The question of whether Caland formations should be classified as primary or secondary (called "*the* basic problem" by Nussbaum himself in 1976: 101) has been a crucial one ever since, as it is rarely possible to decide on the derivational base using morphological criteria. However, while the path described in (13) is most likely the ultimate source of the morphological behavior, Caland systems can also develop without attested root nouns. This suggests a reanalysis of the derivational base as being not a (non-overtly marked) nominal formation, but the root itself, giving way to the deradical derivational principle of the system.[19]

Several attempts have been made to trace the diachronic development of PC lexicalization even further back in time. Jasanoff (1978, 2004) discusses the integration and subsequent extension of Indo-European "*ē*-verbs" in the Caland system (see Table 9 and Sections 3.2 and 3.3 below) and the formation of stative-intransitive verbs more generally, arguing for a nominal, (de-)instrumental origin of the *ē*-suffix (cf. Grestenberger 2023 with refs.).

[18] Note that except for *-*i*- these are the suffixes illustrated in Table 2.

[19] The origin of the Caland substitution rule, its connection to the independently observed *-*o*- → *-*i*- replacement rule, and the original function and distribution of the latter are discussed in, e.g., Schindler 1980; Nussbaum 1999, 2022; Rasmussen 2002; Balles 2006: 269–287, 2009; Pinault 2018; Grestenberger 2021. For reasons of space, we do not discuss this rule here.

As mentioned above, Balles (2006, 2009) has argued that PIE did not have a categorial noun/adjective distinction and that PCs were encoded either as verbal (cf. 4) or through underspecified PC nominals as discussed above in Section 1. The Caland system evolved only after the need to differentiate the adjectival from the abstract reading of these PC nominals arose, which according to her also explains why Caland adjectives "are semantically primary, but formally secondary, namely derived" (Balles 2009: 10–11). In other words, "Caland's Law" is an epiphenomenon of the emergence of adjectives as a distinct word class, which was only in its beginnings when the Anatolian branch left the family. The *i*-stem of the *cvi*-construction in Table 3 would then be a relic from the time before this development started.

While in Balles' analysis PIE could encode adjectival semantics both via a verbal and nominal strategy (2006: 271) and the CS is the result of the emergence of an adjectival word class from the nominal strategy, Bozzone (2016) argues that the CS is a relic of an older *verbal* strategy of lexicalizing PCs. She argues that Caland adjectives were originally inflected and behaved syntactically like verbs, based on a comparison with Japanese PCs. According to Bozzone, PC-associated verbal formations were specifically root aorists with fientive semantics, remnants of which can be found in some Vedic aorist participles. These verbs were thus effectively constructions of type (4c) above, which were productive in pre-PIE but became obsolete in the history of PIE due to a shift from a verb-like adjective class to a noun-like one and correspondingly to differences in adjectival predication. Before this shift, pre-PIE could encode adjectival meanings either through Caland root verbs or via predicative instrumentals of nouns that were *not* part of the CS. After the shift, Caland roots were able to form both root verbs and root nouns, whose predicatively used instrumentals became verbalized and led to the aforementioned stative *ē*-verbs. The root verbs, on the other hand, were specifically associated with perfective aspect (aorists). This would explain why the competing predicatively used instrumentals became associated with *present* stem formation and would fit the typologically common pattern of encoding fientive/CoS semantics as verbal and stative semantics as nominal. Finally, the proposed resulting "switch adjective system" is, per Bozzone, typical of the last stage of a shift from verb-like to noun-like adjectives.

Though innovative, both accounts are problematic for several reasons. Balles' account wrongly dismisses the Anatolian evidence for an old (inherited) class of primary adjectives and at the same time overestimates the age of adjectival *i*-stems both in Anatolian and Indo-Iranian, which are likely to be secondary, independent innovations (Grestenberger 2009: 8–10, 2014: 99; Pinault 2018: 316).

Bozzone, on the other hand, places too much emphasis on Vedic participial formations in her reconstruction of pre-PIE PC aorists, many of which are usually analyzed as Caland *adjectives*, e.g., Ved. *bṛh-ánt-* 'high', which may have been originally denominal to a root noun (cf. Table 2 and the discussion there; see also Lowe 2012, 2015: 283–294). Bozzone takes them as fossilized participles from root aorists without providing independent evidence for *finite* forms of such aorists.

Despite these issues, these accounts provide the most detailed extant proposals for how originally nominal (instrumental) forms came to be integrated into the verbal system in the attested IE languages and thus constitute an important starting point for further studies, even though the internal reconstruction of the prehistory of the Caland system must necessarily remain speculative to some extent. One of the main problems that emerges from this work is the nature of primary verb formation of PC roots vs. non-adjectival CoS roots (cf. Section 2.2), especially with respect to what types of primary verbs these two classes could form (e.g., presents vs. aorists) and to what extent the morphological strategies for forming such verbs overlapped (see Rau 2009a, Rau 2013, this volume).

Before we fully turn to the verbal side of the system, it is moreover important to stress that while numerous Caland roots correspond to Dixon's (1982) property concepts in Table 1 (anticipated by Watkins 1971 and Nussbaum 1976 and subsequently shown by Balles 2006, 2009 and Rau 2009a), "it is not true that a CS guarantees an adjectival root and *vice versa*" (Nussbaum 2022: 207). That is, not every root with a PC meaning necessarily forms a Caland system (the most prominent example being PIE **néu̯o-* 'new') and not every root displaying a Caland system necessarily has PC meaning from the beginning, e.g., **pleh$_1$-* 'fill (up)', whose meaning is not prototypically adjectival but whose derivatives across the family show that a securely reconstructable system has developed over time (see Nussbaum 2022: 208, fn. 7 for a possible path of this development). As already mentioned in Section 2.2, the nominal as well as the verbal morphology involved is not *exclusively* found in PC/Caland roots, and consequently, some derivatives from roots that entail a CoS are discussed under the label "Caland" in the literature as well (e.g., in Rau 2009a, Rau 2009a, Rau 2013, this volume; Bozzone 2016; Majer 2017).

A frequently quoted example is **b^{h}eu̯d^h-* 'awake, become alert' where a (possibly) primary stative verb next to a **ro*-adjective is attested. Table 8 shows a comparison of the respective derivatives with those of a (canonical) PC root in Baltic and Slavic. Pairs like these deserve future work, especially concerning criteria for a potential differentiation of the two types both for specific languages and cross-linguistically.

Table 8: Shared morphology in CoS vs. PC root derivatives in Baltic and Slavic (ex. from Majer 2017)

Suffix		a. CoS root *b^heu̯d^h-		b. PC root *$h_1$reu̯d^h-
*-*ro*-	Sl.	*bŭdrŭ* 'alert, vigilant'	Sl.	*rŭdrŭ* 'red'
	Lith.	*budrùs* 'id.'	Lith.	–
*-*ē*-	Sl.	*bŭděti*, *bŭdi*- 'be alert'	Sl.	*rŭděti*, *rŭdi*- 'become red'
	Lith.	*budė́ti*, *bùdi* 'id.'	Lith.	?[a]

[a]Watkins 1971: 64 gives Lith. *rudė́ti* 'become reddish' as cognate (compare also Fraenkel 1962–1965: 745 s.v. *rùdas*), but according to ALEW s.v. *rūdìs* and LKŽ s.v. *rūdė́ti*, this verb does not show an *i*-present but a present in -*ė́ja*, which is expected of productive denominatives. The shape of the root vowel, which is long (i.e., *recte* ***rū****dė́ti*, as quoted by Majer 2017: 34), and the present point to a synchronic denominative from *rūdìs* (so ALEW). However, LIV[2] 508 gives a first singular *rūdžiù*, and LKŽ also indicates a present variant in -*i*. Furthermore, Latvian *rudêt* shows the historically expected root vowel quantity. Therefore, it is possible that the present in -*ė́ja* is a later formation. See also Pakerys 2006: 67, with fn. 3.

In other words, although PC and eventive roots seem to be able to diachronically switch class, they need to be carefully separated synchronically in order to make the argument that PCs were originally encoded as verbs of a particular morphological class (or classes).

3.2 Verbal periphrasis in the Caland system

While the morphosemantics of the Caland system in the nominal domain are increasingly well understood thanks to Nussbaum (1976) and much subsequent work on various branches of Indo-European (e.g., Meiser 1993; Nussbaum 1999; Balles 2006, 2009; Rau 2009a; Dell'Oro 2015; Bozzone 2016; Majer 2017, 2021), the verbal domain has received less attention so far. Watkins (1971) was one of the first to point out the close association between Caland adjectives and stative verbs in (*)-*ē*-, arguing that such verbs were derived from adjectives via the same substitution principle as in the nominal domain. That is, the verbal suffix -*ē*- does not select the full adjectival stem but the root, with apparent deletion of the adjectival suffix. This analysis was adopted by Nussbaum (1976) and subsequently investigated further, amongst others, by Jasanoff (1978) and (2004) (see (8a-i) for some examples). The ultimate source of the suffix is again to be sought in root nouns,[20] but unlike in (13), the derivational base was not the noun stem, but a

[20]The alternative account which reconstructs *-*(e)h_1*- as primary verbal stem-forming suffix (Harðarson 1998; LIV[2] 25) is problematic for several reasons; cf. Jasanoff 2004, García Ramón

case form, namely an instrumental in *-eh_1 (> -*ē*). Such instrumentals seem to have been able to become the basis for “decasuative” verbal derivation[21] using the denominal verbalizing suffix *-*i̯e/o-*, whence the sequence *-eh_1-*i̯e/o-* found in present stem formations across the IE languages (Watkins 1971, Rasmussen 1993, Harðarson 1998, Jasanoff 2004, Yakubovich 2014; cf. fn. 20), for example, in the Latin 2nd conjugation presents of the type *clār-**ē**-re* ‘be(come) clear’ (*clārus* ‘clear’) or *sen-**ē**-re* ‘be old’ (*senex* ‘old’) etc.; cf. Table 11 below. As these examples show, these verbs are synchronically deadjectival rather than built on a root noun and use the same Caland replacement pattern discussed in Section 3.1.

The second relevant strategy is illustrated in Table 9 and involved periphrastic constructions using finite light verbs or auxiliaries, similar to the *cvi* constructions in Table 3.

Table 9: Light verb constructions with the root noun instrumental *h_1rud^h-$éh_1$ ‘with redness’

Base	Light verb/AUX	Meaning	Semantic type
*h_1rud^h-$éh_1$	h_1es- ‘be’	‘be red’	stative
*h_1rud^h-$éh_1$	b^huH- ‘become’	‘become red’	fientive
*h_1rud^h-$éh_1$	d^heh_1- ‘put’, ‘make’	‘make red’	factitive

Univerbations of these constructions are found in the individual languages and have given rise to productive fientive and/or factitive verb formation types, e.g., the Latin types *rub**e**-fīō* ‘become red’ (< … *b^huH- ‘become’) and *rub**e**-faciō* ‘make red, redden’ (< … *d^heh_1- ‘put, place’), but there are also remnants of the older analytic construction, e.g., Vedic *gúh**ā** bhū-/kr̥-* ‘become/make hidden/with hiding’ (*gúh-* ‘hiding place’, -*ā* INSTR.SG) and Latin analytic *facit ar**ē*** ‘make hot/with heat’ besides the more common *ār**ĕ̄**-faciō* (Hahn 1947; Weiss 2020: 138, fn. 18). This construction has played a crucial role in the debate on the diachronic integration of PC roots into the IE verbal system (e.g., Schindler 1980; Jasanoff 2004; Balles 2006, 2009; Bozzone 2016; see also MERRITT, this volume). However, since these (univerbated) light verb constructions were arguably denominative in origin and thereby morphologically secondary (note here the discrepancy to the synchronic primary status), it remained unclear whether it was possible to derive verbs with

2014: 156, Grestenberger 2023: 20–21 for discussion and criticism.

[21] See PINAULT and GARNIER, this volume, on decasuative verbs, and Fortson 2020 for an evaluation of decasuative derivation in IE and in general.

primary verbal suffixes from Caland roots in the proto-language. Important evidence for this question was added to the Caland canon only relatively recently by Rau (2009a, 2013), who argues for a systematic association of primary verbs like full-grade thematic presents, nasal-infix formations and iterative-causatives in **-éi̯e/o-* with PC roots (see also the discussion of Bozzone 2016's account in Section 3.1).

Still, some scholars argue that these verbs must be later, parallel innovations of the individual languages and the only strategy of the proto-language to derive verbs from PCs was via a nominal instrumental. Given that the root-based derivational pattern could have been triggered by the reanalysis of the derivational base from a nominal with a zero suffix to an uncategorized root as outlined above (Section 3.1), this potential diachronic development could also have given rise to the derivation of primary verbs from Caland roots. Thus, more research into the verbal correlates of Caland adjectives/PC roots is needed: Although productive means of deriving deadjectival verbs exist in all older Indo-European languages, these have for the most part not been the subject of systematic study and analysis from the perspective of comparative reconstruction or theoretical linguistics, with the exceptions mentioned above. There are several reasons for this fact: On the one hand, verbal secondary derivation is in general less well studied than nominal derivation or primary verb formation (cf. Villanueva Svensson, this volume) and on the other hand, the synchronic productivity of various strategies makes it difficult to distinguish forms that are *einzelsprachlich* from inherited strategies. Within the Caland system, the morphological "replacement" rule of the adjectival stem-forming suffix usually makes it impossible to distinguish primary from secondary formations morphologically, requiring the development of additional diagnostics for establishing derivational directionality that hold cross-linguistically (cf. Marchand 1964; Arad 2003; Balteiro 2007; Lohmann 2017; Grestenberger & Kastner 2022) in order to distinguish between synchronic and diachronic patterns of derivation, especially in cases in which erstwhile adjectival morphology has become reanalyzed as verbal stem-forming morphology.

In the next sections, we briefly discuss some patterns with potential claim to antiquity and also address the issue of voice morphology in deadjectival verbs, which has not been studied systematically so far.

3.3 Deadjectival verbs, voice, and the factitive-fientive alternation

As we saw in Section 2, there is a close derivational link between CoS verbs and primary/property concept adjectives (Hale & Keyser 1998, 2002; Harley 2005, 2011; Koontz-Garboden 2005; Francez & Koontz-Garboden 2017, etc.), and this has also been shown to be the case in the older Indo-European languages (Rau 2013; Yakubovich 2014; Sasseville 2021). This raises the question of the diachronic relationship between deadjectival CoS verbs and other types of statives and CoS verbs (e.g., of the causative alternation, and compare also Table 8), as well as their interaction with voice marking. Thus, in Ancient Greek fientives and factitives (recall from Section 2.2 that these are our terms for intransitive vs. transitive CoS verbs from adjectival roots) are usually formed from the same verbal stem, but alternate between active endings in the transitive-factitive and nonactive or "middle" endings in the intransitive-fientive forms, independent of the verbalizer that is used, cf. Table 10.[22]

Table 10: Alternating deadjectival CoS verbs in Ancient Greek

Adj.	Act. factitive	Nonact. fientive
leukós	*leukó-ō*	*leukóo-mai*
'bright, white'	'make white, whiten'	'turn white'
orthós	*orthó-ō*	*orthóo-mai*
'straight'	'set up straight, straighten'	'sit up, become straight'
kratús	*kratū́n-ō*	*kratū́no-mai*
'strong'	'make strong, strengthen'	'become strong'
barús	*barū́n-ō*	*barū́no-mai*
'heavy'	'make heavy, weigh down'	'be(come) heavy'
thermós	*thermaín-ō*	*thermaíno-mai*
'warm, hot'	'make hot'	'warm up, become hot'
leukós	*leukaín-ō*	*leukaíno-mai*
'bright, white'	'make white, whiten'	'turn white'

However, there is a second pattern in which both the fientive and the factitive alternant are morphologically active and which has not been systematically studied in the older IE languages. There are two variants of this pattern: One in which the fientive and the factitive alternants share the same stem (also called the "labile" pattern) and one in which they form different verbal stems. The "labile" pattern is sometimes discussed under the label "unmarked anticausatives"

[22]The diachrony of these verb classes and in particular the extension of the deadjectival verbalizers *-óe/o-*, *-ū́ne/o-*, *-aíne/o-* to new adjectival bases through reanalysis of the adjectival stem-forming morpheme as part of the verbalizer is discussed in Tucker 1981, 1990; Rau 2009a: 115–120, 152–155; Villanueva Svensson 2024; on the theoretical aspects of this reanalysis cf. Grestenberger 2022, 2023.

in the literature (e.g., Schäfer 2008; Heidinger 2010; Alexiadou et al. 2015) because it is part of a broader pattern by which anticausatives can be either marked (e.g., by a reflexive or detransitivizing affix or particle) or unmarked with respect to the transitive-causative alternant. In Modern Greek, the unmarked strategy is in particular associated with deadjectival CoS verbs, as noted by Alexiadou & Anagnostopoulou (2004: 125), who cite *asprizo* 'whiten', *kokinizo* 'redden', *mavrizo* 'blacken', *katharizo* 'clean', *stroggilevo* 'round', *klino* 'close', *anigo* 'open', *plateno* 'widen', *stegnono* 'dry', *stenevo* 'tighten', *skureno* 'darken', *kathistero* 'delay', *alazo* 'change' and *ksepagono* 'defreeze' as examples of verbs in which both the causative and the anticausative variants obligatorily take the active endings, i.e., verbs which do not alternate.[23] This pattern arose during the Hellenistic period and seems to have begun among non-deadjectival anticausative/CoS verbs, judging by the examples discussed by Lavidas (2009: 112–116), who links the rise of unmarked anticausatives to the more general restructuring of the distribution of active vs. nonactive morphology that took place in Greek around that time (though the details, especially with respect to the extension to deadjectival verbs, require further elaboration). A similar rise of morphologically active intransitivizations among alternating CoS verbs is discussed by Gianollo (2014) for late Latin. However, it remains unclear to what extent such a "labile" pattern existed in the older IE languages and whether it can be reconstructed for PIE.

In light of the diachrony of the thematic active endings, which are formally related to the reconstructed *middle* set of endings and explicitly treated as reanalyzed former middle endings in Jasanoff's "*h_2e-conjugation" framework (Jasanoff 1978, 2003, 2004, 2017a, 2019, 2023; cf. also Watkins 1969; Hollifield 1977; Villanueva Svensson 2003, 2021 on the origin of the thematic conjugation), apparently "labile" forms in which the active-marked alternant appears in intransitive contexts could also be seen as remnants of a stage in which the intransitive-fientive alternant took the nonactive endings as well as a different verbal stem formant than the transitive-factitive alternant (see Meiser 1993, Luraghi 2012 for different versions of this view) – that is, verbs that failed to undergo the "formal update" to the synchronically productive middle endings. This is, for example, the standard explanation since (at least) Watkins 1969 for why some *active* thematic aorists such as AG *édrakon* 'saw' and *étraphon* 'grew

[23]There is also a class in which the anticausative alternant can take either the active or the nonactive endings, that is, it can form both unmarked and marked anticausatives. The difference seems to be partially conditioned by the animacy of the subject in the anticausative variant, though the degree of spontaneity or completion of the change of state have also been adduced as possible factors. See Alexiadou & Anagnostopoulou 2004, 2013; Alexiadou et al. 2015: 88–90 for further discussion.

(strong)' have formally *middle* presents (*dérkomai* and *tréphomai*, respectively; see also Rau 2013, 2016 on this type of development). A "labile" stage at the PIE level is therefore unlikely.

The second relevant pattern, in which both the transitive and the intransitive alternant are formally active, but built from different verbal stems, is better studied, especially in Latin and Hittite. In Latin, the suffix *-*ē*- that is also found in the Ancient Greek passive aorist and a number of other verbal categories across the IE languages (see Section 3.2) gave rise to a subclass of the Latin 2nd conjugation presents in the form of the stative(/fientive) verbs in -*ē-re* which obligatorily take the active endings. In Old Latin, their transitive-factitive alternants are active 1st conjugation presents, as illustrated in Table 11 (later replaced by the *rubefaciō*-type, cf. Section 3.2). The -*ē*-stem was moreover also used as the base for (likewise originally active-marked) fientives in -*ē-sce-re*.

Table 11: Deadjectival statives, fientives, and factitives in (Old) Latin (Watkins 1971: 47)

Factitive	Stative	Fientive	Adjective
*clār-**ā**-re*	*clār-**ē**-re*	*clār-**ē**-**sce**-re*	*clār-us, -a, -um*
'make clear'	'be clear'	'become clear'	'clear'
*-alb-**ā**-re*	*alb-**ē**-re*	*alb-**ē**-**sce**-re*	*alb-us, -a, -um*
'make white'	'be white'	'become white'	'bright, white'
*-nigr-**ā**-re*	*nigr-**ē**-re*	*nigr-**ē**-**sce**-re*	*niger, -ra, -rum*
'make black'	'be black'	'become black'	'dark, black'
*liqu-**ā**-re*	*liqu-**ē**-re*	*liqu-**ē**-**sce**-re*	*liqu-idus, -a, -um*; *lī̆qu-ēns*
'make fluid'	'be fluid'	'become fluid'	'fluid, liquid'

The alternation in Table 11 could be taken to suggest that -*ē*- and its factitive "alternant" -*ā*- are different "flavors" of the verbalizer *v* in the sense of Folli & Harley (2005, 2007); Harley (2013, 2017) associated with stative (BE) and fientive (BECOME) inner aspect (*Aktionsart*) on the one hand, and factitive/agentive (DO) on the other. In other words, the Latin alternation would be the synthetic equivalent of the *analytic* deadjectival factitive-fientive alternation discussed in Sections 1.2 and 3.2, with which it is also connected etymologically via the "stative" stem formant *-*ē*-.

In Ancient Greek, the cognate suffix -*ē*- (and its allomorph -*thē*-) is restricted to the perfective stem, unlike in Latin. However, the original use in Greek is for the most part that of a stative/fientive verbal stem-forming suffix (similar to the Latin examples in Table 11) rather than a Voice marker. Furthermore, it obligatorily takes the active set of endings, again like in Latin. Thus *-*ē*- is associated with the formation of "unmarked anticausatives" both in Latin and in Ancient Greek. In Ancient Greek, however, it is not specifically associated with PC roots/roots that form primary adjectives (though a few such forms are also found), but with CoS verbs more broadly. Examples are given in Table 12.

Table 12: Homeric CoS *ē*-aorists

*e-pág-**ē**-n* PST-become.fixed-**V.PFV**-1SG.PST.ACT	'became fixed, coagulated'
*e-mán-**ē**-n* PST-rage-**V.PFV**-1SG.PST.ACT	'went mad, became enraged'
*e-ág-**ē**-n* PST-break-**V.PFV**-1SG.PST.ACT	'broke' (intr.)
*e-tárp-**ē**-n* PST-enjoy-**V.PFV**-1SG.PST.ACT	'enjoyed, delighted in'
*e-phán-**ē**-n* PST-appear-**V.PFV**-1SG.PST.ACT	'appeared'

Crucially, the transitive factitive/causative alternants corresponding to these fientive/inchoative aorists use different stem formants (and the active endings) both in the imperfective (present) and the perfective (aorist) stem, cf. Table 13.

Table 13: Ancient Greek intransitive CoS *ē*-aorists and their transitive presents & aorists; stem formant = **bold**

Intr. aor.	Meaning	Tr. pres.	Tr. aor	Meaning
*e-ág-**ē**-n*	'broke'	*ág-**nū**-mi*	*é-ak-**s**-a*	'break/broke'
*e-tárp-**ē**-n*	'enjoyed '	*térp-ō*	*é-terp-**s**-a*	'cause(d) joy'
*e-tráph-**ē**-n*	'grew up '	*tréph-ō*	*é-trep-**s**-a*	'cause(d) to grow'
*e-pág-**ē**-n*	'became fixed'	*pḗg-**nū**-mi*	*é-pēk-**s**-a*	'fix/ed'
*e-rrág-**ē**-n*	'break'	*rhḗg-**nū**-mi*	*e-rrēk-**s**-a*	'break/broke'

The same pattern – the intransitive-fientive and the transitive-factitive are formed from different stems, but both take the active endings – is found in Hittite, suggesting that this pattern is indeed of some antiquity (see Sasseville, this volume). Table 14 gives some examples from Hittite, in which the fientive is formed with the verbal stem formant *-ēšš-*, which has been etymologically linked to Latin statives in *-ē-re* and their associated fientives in *-ē-sce-re* at least since Watkins 1971. The corresponding factitive uses a nasal suffix *-nu-* that has been historically linked to nasal infix-presents from adjectival *u*-stems (Koch 1973, 1980; Sasseville 2021: 482–487, this volume). Synchronically, however, these are formed to adjectives with adjectival stem formants other than *-u-* as well, as Table 14 shows.

Table 14: Hittite deadjectival fientives & factitives (Rau 2009a: 112–113)

Adj.		Fientive	Factitive	
palḫi-	'broad'	*palḫ-**ēšš**-zi*	*palḫ-(a)**nu**-zi*	'broaden'
daluki-	'long'	*daluk-**ēšš**-zi*	*daluk-**nu**-zi*	'lengthen'
daššu-	'strong'	*daššau̯-**ēšš**-zi*	*daš(ša)-**nu**-zi*	'strengthen'
parkui-	'pure'	*parku-**ēšš**-zi*	*parku-**nu**-zi*	'become pure/purify'

Crucially, both the fientive and the factitive stem take the active *mi*-endings.[24] Together with the Latin and Ancient Greek evidence discussed above, this suggests that at least certain classes of stative and fientive deadjectival CoS verbs originally took the active **mi*-conjugation endings and were hence "unmarked anticausatives" in the sense of Schäfer (2008), as argued by Grestenberger (2023) based primarily on the Ancient Greek facts. Since there is a wealth of evidence that suggests that intransitive CoS verbs from result/eventive roots were originally associated with the *h_2*e*-conjugation or "proto-middle" endings (e.g., the "stative-intransitive" aorists of Jasanoff 2003, the intransitive nasal stems discussed by Gorbachov 2007 and Rau, this volume, the simple thematic presents of Rau 2013, 2016, among others), this could point to an original situation in which deadjectival stative and CoS verbs canonically took the active (**mi*-conjugation) endings, whereas CoS verbs from result/eventive roots took the (proto-)middle (*h_2*e*-conjugation) endings, a distribution that was subsequently given up or restructured in the attested daughter languages. This restructuring resulted in the

[24]The deadjectival factitive *nu*-stems take the *ḫi*-conjugation endings in the Luwic branch of Anatolian, which Sasseville (2021: 486) takes to be the original conjugation based on the parallel with the factitive *-aḫḫ*-type, for which *ḫi*-conjugation endings are reconstructed for Proto-Anatolian. He argues that this is due to both types being derived by conversion, but this is difficult to accept at least for the *nu*-stems, which are not instances of conversion in the usual sense since there is an overt verbal stem formant, *-n(u)-*. Moreover, it is not clear why conversion should automatically lead to *ḫi*-conjugation endings. See also Sasseville, this volume, for further discussion of *nu*-causatives and factitives in Anatolian.

mix of strategies that we see in the attested languages as well as in the final, "hybrid" type which uses both different stems and an alternation in voice marking, as we see especially often in Indo-Iranian pairs of full grade thematic middle fientives/inchoatives besides active nasal infix or *-*e̯ie/o*-factitives/causatives (Rau 2009a, 2013, 2016), cf. Table 15.

Table 15: Middle-marked intransitive, active-marked transitive CoS verbs with concomitant stem change in Vedic (Rau 2016: 358–359)

Intr. middle (full grade thematic)		Tr. active (nasal infix)	
*jáva-**te***	'speed'	*junā́-**ti***	'make speed'
*śóbha-**te***	'appear beautiful'	*śumbhá-**ti***	'make beautiful'
*páva-**te***	'become pure, purify oneself'	*punā́-**ti***	'purify'

Rau (2016, this volume) argues that the pattern in Table 15 can be reconstructed for alternating CoS verbs from Caland and non-Caland (eventive) roots at least for the inner Indo-European stage and that its further development, especially in terms of the voice marking of the alternants, is particularly relevant for understanding the development of factitive/causative alternation verbs in Indo-Iranian and Ancient Greek. Taken together, these case studies suggest that the interaction of voice marking and stem formation in deadjectival and result CoS verbs in Indo-European deserves further study.

Another deadjectival factitive class that has recently garnered attention is the "*nēwaḫḫi*-type", named after a class of Hittite factitives to adjectival bases formed with a stem formant -*aḫḫ*- < *-*eh*$_2$- and the *ḫi*-conjugation endings (Jasanoff 2003: 139–141; Hoffner & Melchert 2008: 175–176; Sasseville 2015, 2021), cf. Table 16. Formal comparanda of this type have been identified in Ancient Greek (Rau 2009b) and Baltic (Jasanoff 2003: 140–141, 2017b: 200–201; Villanueva Svensson 2020), which, however, diverge semantically: While the Anatolian -*aḫḫ*-verbs are factitives, the Baltic comparanda are iteratives (the Greek evidence is too scarce to be conclusive).

The verbal stem formant *-*eh*$_2$- has been identified with the abstract- and collective-forming *nominal* suffix *-*eh*$_2$- (Sasseville 2015, 2021, followed by, e.g., Villanueva Svensson 2020), suggesting a derivational chain as in (14), in which a deadjectival verb 'renew' is derived from a deadjectival abstract noun 'newness' via zero derivation or "conversion", as argued by Sasseville (primarily based on Lycian, where this pattern is productive).

Table 16: Hittite deadjectival *nēwaḫḫi*-factitives (Hoffner & Melchert 2008: 176)

Adjective		Factitive	
ḫappinant-	'rich'	*ḫappin-aḫḫ-*	'make rich'
ikuna-	'cold'	*ikun-aḫḫ-*	'make cold'
nakki-	'important'	*nakkiy-aḫḫ-*	'regard/treat as important'
nēwa-	'new'	*nēw-aḫḫ-*	'make new'

(14) **néu̯-o-* 'new' →
**néu̯-eh$_2$-* 'newness, news' →
**neu̯-eh$_2$-Ø-* 'make new/with newness', 3sg. **neu̯-eh$_2$-Ø-e(i)* 'makes new'

A formal parallel for deriving deadjectival factitives from a nominal rather than adjectival surface form comes from English, cf. *long* – *length-en*, *strong* – *strength-en*, though with overt verbalizing morphology. This type of derivation would then have given rise to the Hittite and Baltic forms of this type, in which *-aḫḫ-*/**-ā-* is synchronically a verbal stem formant and the formal and semantic basis of this class are PC adjectives of various stem types (Hittite) or verbal adjectives (≈ "result statives" from non-deadjectival roots, as in Table 4) in Baltic.[25] However, this depends on whether one accepts that conversion was a productive verbal derivational process in PIE[26] (so also Villanueva Svensson 2021) and requires further explanation of the semantic discrepancy (factitive vs. iterative) and the reason for the association of this type with **h$_2$e*-conjugation endings (for a novel proposal concerning the latter point see Pinault, this volume).[27]

[25] Though synchronically the type is "deverbative", cf. Villanueva Svensson (2020: 392–395, this volume).

[26] Adjectives in English do not participate in conversion, e.g., **a*/**to white/flat/smart* vs. *a*/*to hammer/table/walk*, cf. Borer 2013: 371–378. Acedo-Matellán (2022a) argues that this is because they are structurally complex, consisting of a preposition and a noun. If this turns out to be a systematic gap cross-linguistically, this would speak against the conversion account of Villanueva Svensson (2021). However, it is not yet clear to what extent Borer's observation holds cross-linguistically or (if it does) why such a complex structure should be compatible with overt but not with zero verbalizers (assuming that that is what conversion is; see Grestenberger & Kastner 2022: 47–52 for further arguments).

[27] Jasanoff (2003: 146) tentatively proposes that the factitive value of the type in Hittite is due to secondary transitivization – the formation of an oppositional active factitive to an older intransitive fientive with **h$_2$e*-conjugation endings. However, middle-marked fientive *aḫḫ*-verbs are rare in Hittite, and the proposal has not been taken up further.

3.4 Deadjectival activity verbs and iterativity

Although deadjectival factitive-fientive alternation verbs have received most of the attention in the Indo-Europeanist literature on deadjectival verb formation so far, there are also other verb classes that seem to be semantically and/or morphologically associated with adjectival stems. One class was already briefly mentioned in Section 2.2, namely deadjectival activity verbs like Sp. *holgazanear* 'to act lazily' (*holgazán* 'lazy') or Hebr. *hištaxcen* 'to act arrogantly' (*šaxcan* 'arrogant'), in which the base adjective seems to act as a modifier describing a property of the surface subject. These exist in the older IE languages as well, but have not received much attention so far. Marescotti (2024) discusses this class in Ancient Greek under the label "quality verbs", with examples such as *stōmúllō* 'be talkative, chatter' (*stōmúl-os* 'talkative') and *aióllō* 'shift rapidly back and forth' (*aiól-os* 'quick, nimble'). In Latin, examples include *grātor* 'rejoice' (*grāt-us* 'pleasant, agreeable') and *misereor* 'feel pity for' (*miser* 'wretched, pitiful'), though with a slightly different semantic relationship between the base and the derived verb. This class tends to be derived from morphologically complex adjectives and shares semantic and morphological similarities with denominal verbs from animate agent-like bases ("pseudo-agent verbs") such as AG *basileú-ō* 'be king, rule over' (*basileús*), *tektaín-omai* 'be a carpenter, work as a carpenter' (*téktōn* 'carpenter') and Latin *ancillor* 'serve as a maid' (*ancilla* 'maid, female servant'), *interpretor* 'explain, interpret' (*interpres*, *-pret-*) 'intermediary'), *philosophor* 'act like a philosopher, be philosophical' (*philosophus* 'philosopher'), etc.,[28] including the ambiguity between a state and an activity reading.

A related class we want to briefly discuss here consists of verbs that are usually classified as iteratives or pluractionals and describe a plurality of events. Specifically, these verbs are treated as event-internal pluractionals, in which a plurality of subevents is grouped into a single event (e.g., Tovena & Kihm 2008; Tovena 2010; Greenberg 2010). Because they are synchronically derived from (non-PC) roots or verbal stems, they are usually called "deverbal" or "deverbative", though morphologically they are often root-derived (cf. fn. 4 and 12 above). However, from the point of view of their derivational morphemes, such iteratives (used here as a cover term) often seem to be associated with *adjectival* morphology at least diachronically, specifically with "verbal adjectives" from eventive roots rather than PC roots.[29] This pattern is attested in Latin, for example, which has a

[28] See for example Flobert (1975: 666–674), who treats both the deadjectival and the denominal verbs of this types as "predicative" verbs. On "pseudo-agent verbs" see also Xu et al. 2007: 137–139; Bleotu 2019: 163–168; Marescotti & Grestenberger 2026.

[29] Though Rau (2009a, 2013) also discusses iteratives that are associated with the Caland system.

synchronically deverbal class of verbs that forms "repetitive verbs" using a suffix -*tā*-. The -*t*- is generally agreed to be identical to the perfective passive participial suffix at least synchronically (Weiss 2020: 424),[30] -*ā*- marks conjugation class I, cf. ex. a. in Table 17. This class forms event-internal pluractionals, whereas the "frequentatives" in -*it*-*ā*- in ex. b., which are also based on what seems to be the participial -*t*-stem (cf. -*t*-*us*), form multiple-event pluractionals.

Table 17: Latin repetitives and frequentatives

a. Latin repetitives			
cap-e-re	'seize'	*cap-**t**-ā-re*	'capture'
hab-ē-re	'have'	*habi-**t**-ā-re*	'inhabit'
b. Latin frequentatives			
ag-e-re	'do'	*ac-**t**-**it**-ā-re*	'do repeatedly'
can-e-re	'sing'	*can-**t**-**it**-ā-re*	'sing repeatedly'

A second pertinent example is found in Germanic, for example in Old High German (OHG), which has a class of synchronically deverbal iteratives in -*ar*-. These inflect as weak class II verbs independently of the conjugational class of the base, cf. Table 18.

Table 18: OHG -*ar*-iteratives

Base		Iterative	
wach-ēn	'be awake'	*wach-**ar**-ōn*	'to stand guard'
wīg-an	'fight'	*weig-**ar**-ōn*	'refuse, be obstinate'
gang-an	'go'	*gang-**ar**-ōn*	'walk around'

[30] Diachronically, the situation is less clear – this class may have originated in denominal verbs based on abstract nouns to *s*-desideratives, cf. Nussbaum 2021b: 18 & fn. 49 (for an opposing view cf. de Vaan 2012). The repetitives of the type *ag-it-ā-re* (< **ag-et*-) 'shake, agitate' (*ag-ere* 'drive'), which are not based on the synchronic participial stem, on the other hand, seem to be derived from agent(ive) nouns of the type *eques*, *equit*- 'horserider', whence *equit-āre* 'to be a rider, to ride' (see Nussbaum 2017: 260–263 on the functions of nouns in *-*(e)t*-). In that case, this class of repetitives would be derivationally similar to the "activity verbalizations" discussed in the main text above. Cf. also Weiss 2020: 424–425 for a recent discussion of the origin of the Latin repetitives and frequentatives.

The handbooks locate the source of the *-ar*-suffix in adjectival *r*-stems such as *wach-ar* 'awake', *weig-ar* 'obstinate', etc. (e.g., Wilmanns 1896: 92–95), which are usually formed to result roots, though comparatives in *-(e)r-* may also have played a role.

The Baltic continuants of the "*nēwaḫḫi*-type" have already been mentioned in this context in the previous section (Section 3.3). In Baltic, this class forms deverbal iteratives using a suffix *-*ā*-. The diachronic derivational basis of these stems seems to consist of verbal adjectives in *-*to*- (cognate with the Latin -*t*-participles) and *-*o*-. This is implied by the root *o*-grade and the remnants of adjectival morphology in the iteratives while the origin of the *-*ā*-verbalizer is debated (cf. Section 3.3). Table 19 shows some examples from Lithuanian.

Table 19: Lithuanian iteratives (citation form = 1SG)

Iterative		Base verb		Verbal adjective	
***glóst**-au*	'stroke'	*glódž-iu*	'smooth, polish'	***glós-t**-as*	'smooth'
***laik**-aũ*	'hold'	*liek-mì*	'remain'	*-**laik**-as*	'remaining'

What these iterative verb classes have in common is that 1) they inflect according to the synchronically productive inflectional class for forming denominal verbs (e.g., the first conjugation in Latin), independent of the inflectional class of the base, 2) they are atelic iterative/pluractional verbs, and 3) their derivational morphology can plausibly be linked to that of root-derived verbal adjectives at least diachronically. Unlike deadjectival verbs from property concept or "adjectival" roots, which usually form degree achievement verbs of the factitive-fientive alternation (Section 3.3), these verbs are formed to adjectives from eventive roots which are not typically gradable or scalar. The derived iterative verbs themselves are atelic activity verbs rather than achievements (cf. also Tovena & Kihm 2008; Tovena 2010; Grestenberger & Kallulli 2019 on pluractionality and the semantics of "verbal diminutives").

As this brief survey shows, different types of adjectives give rise to different types of deadjectival verbs, and this is reflected both in the morphology and the semantics of the resulting verb classes. However, more work is needed to explain why verbal adjectives become the input to verbal derivation in the first place, how iterativity arises compositionally from such derived verbs, and how they are connected to other types of deadjectival activity verbs.

3.5 Summary

In this section, we have briefly summarized the current state-of-the-art on the Caland system, the lexicalization of PCs in the older Indo-European languages, and the debate around the nature and origin of the CS. We have also provided a survey of different types of periphrastic and synthetic verb forms associated with deadjectival verb formation in these languages, focusing on the factitive-fientive alternation and the formation of different types of statives. This discussion is far from exhaustive, and we refer to the articles in this volume for a more in-depth discussion – for individual branches in particular the contributions in part I.

In the next section, we provide a brief outline the structure of the volume and summarize the contributions.

4 Contributions

Because of the intentionally broad definition of "deadjectival" we have proposed in Section 2.1 and the "categorial instability" that is cross-linguistically observable with respect to the lexicalization of PCs, the contributions collected in this volume cover not only the factitive-fientive alternation, but also deadjectival verbs in which the semantic derivational basis appears to diverge from the morphological basis, as in the famous case of the "*nēwaḫḫi*-type" (cf. Section 3.3 and the contribution by Pinault). This volume is therefore also intended to connect to the existing literature on denominal verb formation in Indo-European more generally, which usually treats verbal derivation from both adjectival and substantival nominal bases (e.g., Tucker 1988, 1990; Insler 1997), as for instance in the contributions by Merritt and Pinault, as well as to more recent work on factitive-fientive alternations related to the Caland system and the discussion concerning the potential deadjectival origin of the PIE *u*-presents and simple thematic presents (Villanueva Svensson 2021; Jasanoff 2023).

The contributions are divided into three parts: Part I contains contributions that aim to fill empirical gaps in languages or subbranches of Indo-European in which deadjectival verbs are understudied, namely Boldt for Old Norse, Höfler & Stifter for Celtic, Sasseville for Anatolian, and Villanueva Svensson for Balto-Slavic.

In "To be or to become, that is the question: The morphosemantics of the weak verbs formed to color adjectives in Old West Norse", Ramón Boldt discusses the morphology and semantics of the verbal derivatives of eight Basic Color Terms (BCTs) in Old West Norse, thus filling a gap in the literature on Old Norse verbal derivation. His examination of the derived weak verbs shows that they display the semantics expected of deadjectival formations, namely factitive and fientive/anticausative meanings. However, in some cases the translations offered by the dictionaries lead to unclear or even contradictory readings both concerning their sense and the mapping between morphology and semantics. Therefore, the verbs in question require a new philological analysis of the source texts. Boldt argues for determining the semantics as precisely as possible for these problematic cases in order to identify the original semantics of the different verb classes and their diachronic development. He also discusses their associated stem formation types.

Stefan Höfler and David Stifter's article "Caland verbs in Celtic" discusses deadjectival verbs in an Indo-European branch that has not yet been the subject of a comprehensive treatment within Caland-related studies, especially regarding verbal derivation. The authors show that at least 20 verbal stems associated with the Caland system can be found in Celtic. While half of them have identifiable cognates in other branches and are therefore likely to be inherited, the other half lack extra-Celtic cognates. However, within those probable inner-Celtic formations, based on the evidence of Old Irish and Middle Welsh the authors identify verb types that are clearly related to the Caland system, such as nasal-infix presents and iterative-causatives as well as a stative and a simple thematic present. They conclude that this evidence speaks in favor of a mild productivity of the root-deriving principle of the Caland system in Proto-Celtic.

David Sasseville discusses the development of the denominal/deadjectival morpheme -*nu*- in the Anatolian branch in his contribution "The relationship between *u*-adjectives and *nu*-causatives in Anatolian". The Anatolian evidence points to a similar derivational pattern as in Old Indic and Ancient Greek, namely the reanalysis of *n*-infixed verbs derived from *u*-adjectives as containing a *nu-suffix*. Though it can be shown that the infixation strategy precedes suffixation, synchronically the forms are often ambiguous regarding their derivational base, i.e., they could be analyzed as being deadjectival or as derived from a root verb (a verb with a zero verbalizer). As Sasseville argues, this situation can lead to yet another process of reanalysis, namely one that gives rise to a "*u*-extension" of roots. While previous research has focused mainly on Hittite, the addition of data from other Anatolian languages provides new insights into the the diachronic interaction of root verbs, *u*-adjectives and *nu*-formations, and demonstrates that the loss of the adjective is not a requirement for a reanalysis of the *u*-extension type. Thus, some verbs in Luwian, Lycian and Palaic can be explained as derived from precisely such *u*-extended roots.

In "Balto-Slavic denominatives from an Indo-European perspective: An update", Miguel Villanueva Svensson provides a timely survey of denominative verb formation in Balto-Slavic, an urgent desideratum both from the perspective of Indo-European and of Balto-Slavic studies. As the author himself states, treatments of this topic are either missing completely or are present only in a short and often outdated way in the literature. This paper serves as an overview that will pave the way towards further research on secondary verb formation in Balto-Slavic. Specifically, Villanueva Svensson argues that despite being notoriously innovative in some regards, the verbal system of Balto-Slavic provides valuable insights into the reconstruction of denominative verb formation in Indo-European. By surveying the different morphological means (i.e., suffixes) attested in Baltic and Slavic, he first identifies the inherited as well as innovated strategies for deriving verbs from nominal bases and then connects them to formations in different branches of the family such as Vedic, Ancient Greek, and, in the case of the *nu*-factitives, Anatolian (cf. the immediately preceding chapter by Sasseville).

The contributions in part II focus on the reconstruction of derivational patterns of deadjectival verbs and property concept roots in Proto-Indo-European and their development into the attested languages.

Romain Garnier discusses "Delocatival adjectives surfacing as participles" in a contribution that addresses the difficult question of categorial affiliation of several participial and adjectival PIE suffixes containing the morpheme **-en-*, which he argues emerged as delocatival formations based on root nouns. This paper thus contributes to understanding how nominal and adjectival suffixes can become integrated into the verbal system and eventually be reanalyzed as verb forms.

Andrew Merritt's paper "De-adjectival and De-"adjectival": Greek κρύπτω 'cover' and Proto-Germanic **χreuđanq̨* 'deck, equip, adorn"' deals with questions of internal derivation and complex derivational chains in the domain of deadjectival verb formation. Merritt shows that within such chains, originally secondary formations such as **-i̯e/o*-presents from adjectives could be reanalyzed as primary verb formations (and even lead to the creation of new roots), which in turn gave rise to new (verbal) adjectives functioning as derivational bases for new secondary formations. Decasuative light verb periphrases with **d^heh_1-* and **b^huH-* to adjectival formations can also serve as bases for the creation of new roots and adjectives. Understanding the diachrony of the Caland system thus provides a tool for linking different synchronic reflexes of various branches (Greek, Tocharian, Germanic, Balto-Slavic) to a derivational "bigger picture".

In "Reassessing the emergence of the PIE **-eh_2*-factitive present type", Georges-Jean Pinault addresses a verbal class that has only recently become the focus of attention in the context of the question whether conversion (the transformation of a nominal stem into a verbal stem, and vice versa, without overt affixation) played a role in verbal derivation in Proto-Indo-European or in specific daughter branches thereof, such as Anatolian. By questioning the exact nature of this process within Proto-Indo-European, the paper discusses three examples for which conversion has been proposed and suggests alternative explanations, partially based on inner-IE derivational processes that are already known, but also by introducing a novel analysis of the *nēwaḫḫi*-type, namely as a reanalyzed dative singular case form of abstract nouns used in predicative function. Contrary to previous explanations of this enigmatic verbal stem class, this approach explains the persistent *ḫi*-inflection of this type in Anatolian.

Jeremy Rau's article "A late Nuclear Proto-Indo-European verbal type: Intransitive thematic nasal-infix present actives" identifies a new verbal derivational class in "late Nuclear PIE" (the ancestor of the remaining IE languages after Anatolian, Tocharian, and Italo-Celtic branched off), namely thematic nasal-infix presents to state-oriented roots. Rau argues that at least some of these presents are "deadjectival" in the sense of being derived from Caland-associated property concept roots, and often associated with the causative alternation (or factitive-fientive alternation) synchronically. He further identifies this intransitive state-oriented type with the "Northern IE" intransitive/anticausative *h_2e-conjugation nasal presents of Gorbachov (2007). This is a significant finding that pushes back the date of reconstruction for this class and at the same time clarifies further aspects of primary verbal derivatives to Indo-European PC/Caland roots and state-oriented roots more generally.

Part III concludes with two contributions that address deadjectival verb formation from a quantitative diachronic perspective (Iacobini & De Rosa) and a theoretical perspective (Tanza-Kamil), thus situating the Caland phenomenon in the broader empirical landscape.

In their article "Diachronic trends in the formation of Italian deadjectival verbs", Claudio Iacobini and Maria Pina De Rosa report the result of a diachronic corpus study on the morphological expression of deadjectival verbs in Italian, based on a dataset of ca. 900 verbs from the 13th to the 20th century. The authors identify three main verb formation processes (suffixation, conversion, parasynthesis), whose productivity varies diachronically, with suffixation becoming the dominant mode of forming deadjectival verbs in the history of Italian. They provide a detailed analysis of the different strategies and their distribution according to adjective type (e.g., simplex vs. morphologically complex) as well as aspectual-semantic class, confirming that the majority of deadjectival verbs are change-of-state verbs. This study therefore provides an important quantitative argument in favor of treating deadjectival/"Caland-associated" verbs as the regular expression (and diachronic source) of the factitive-fientive alternation, as argued in several other articles in this volume for the older IE languages.

In "Deriving the semantics of the Akkadian stative: Perspectives from the adjectival stem", Iris Tanza-Kamil discusses the semantics of the Akkadian stative, a deadjectival construction based on verbal adjectives that express a resultant state from (different types of) CoS roots, but also from unergative, unaccusative, and transitive eventive roots. These are called primary statives, besides which the author identifies "secondary statives" derived from nominal and adjectival stems. She argues that all statives are verbal in terms of their category and resultative in terms of their semantics. This article provides an important typological and theoretical addition to the other contributions in the volume and adduces crucial evidence from a category that is otherwise underrepresented in the discussion of deadjectival verbs, namely verbal adjectives as the input to deadjectival verb formation.

5 Conclusion

In this introductory chapter, we have outlined current research avenues in deadjectival verb formation from the perspective of theoretical linguistics, linguistic typology, and historical Indo-European linguistics. We have discussed the cross-linguistic lexicalization patterns of PCs, the connection between PCs and possessive predication, the nature of lexical categorizers associated with PC roots, and different types of change-of-state verbs. We have also provided a working definition of deadjectival verbs that will hopefully be useful for future investigations of these verb classes.

Our main focus rested on the introduction to and state-of-the-art of research on the "Caland system", that is, the lexicalization and derivational patterns associated with PC roots in (Proto-)Indo-European, followed by a survey of the various verbal derivational patterns in the older IE languages. We have pointed out desiderata as well, such as a stronger engagement with the theoretical and typological literature on PCs cross-linguistically, the need to better understand the difference between root- and stem-based deadjectival verb formation, the role of different types of adjectival morphology in verbal derivation, and the interaction of deadjectival verb formation with various voice and argument structure alternations. Together with the articles collected in this volume, we thus hope that this introduction will serve as a starting point for further work on deadjectival verb formation, the lexicalization of property concepts, and the various ways in which adjectival and verbal morphology interact.

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Abbreviations

Act.	Active
Adj.	Adjective
AG	Ancient Greek
ChSl.	Church Slavic
CoS	change of state
CS	Caland system
Cz.	Czech
Fact.	Factitive
Hitt.	Hittite
IE	Indo-European
Intr.	Intransitive
Lat.	Latin
Lith.	Lithuanian
MG	Modern Greek
Nonact.	Nonactive
OAv.	Old Avestan
OCS	Old Church Slavonic
OHG	Old High German
PC	Property concept
PGmc.	Proto-Germanic
PIE	Proto-Indo-European
PoS	Part of speech
PSl.	Proto-Slavic
Russ.	Russian
Sg.	Singular
Sl.	Slavic
Tr.	Transitive
Ved.	Vedic
YAv.	Young Avestan

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